
LECTURE NOTES

Lecture 5: Innovation

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1 Defining innovations

innovations are changes - these can be...

- technical
- social
- organizational
- scientific

Innovations can include

- New combinations
- New organizational structures
- New methods
- New contents or qualities
- New markets

2 Social innovations

Developing and deploying effective solutions to challenging social and environmental issues — not limited to any one sector, but requiring collaboration across government, business, and nonprofits.

This includes

- Social justice
- Equality
- overall well-being.
- Education
- Healthcare
- Human rights

3 Three big NO's in (social) innovation

- Proven Failure
 - some interventions have no impact at all
 - Often because the actual customer/user/usage case has not been studied carefully
- Negative social outcomes
 - Good intentions can still cause severe harm — immediately or downstream.
- Unethical behavior
 - Kant's golden rule
 - Rawls "veil of ignorance"
 - Utilitarianism
 - Ethical principles of benevolence, non-maleficence & fairness

4 The social impact compass

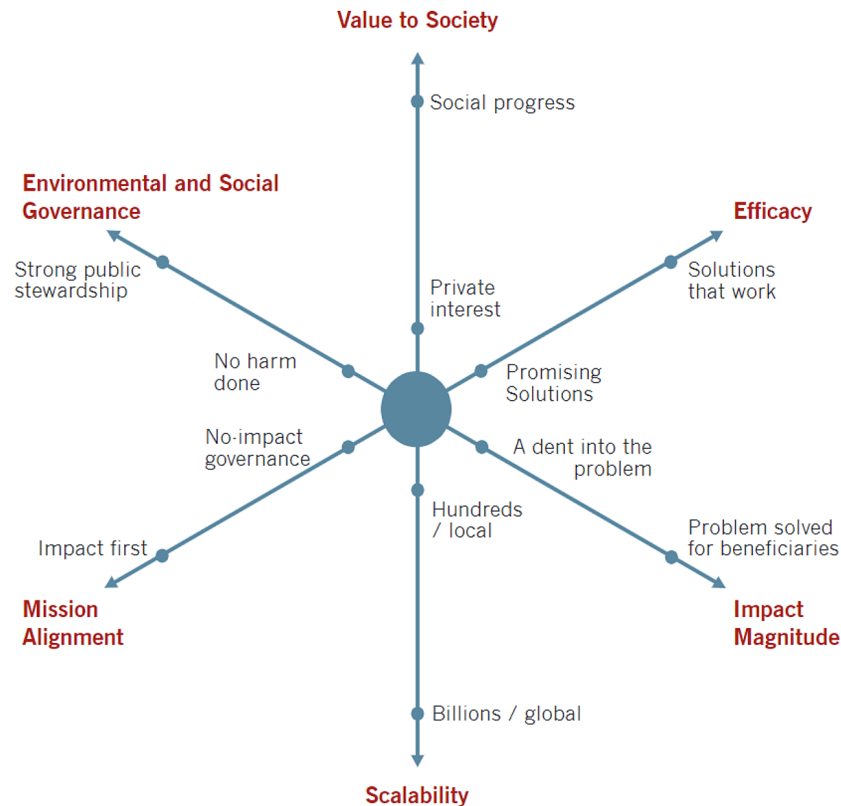


Figure 1:

- Value to society
 - How does the intended outcome deliver societal value?
- Efficacy
 - How certain are we of the effects of the solution?
- Impact magnitude
 - How complete is the solution for each beneficiary?
- Scalability
 - How much of the affected community can this model address?
- Mission alignment
 - To what extent is the organization outcome oriented?
- Environmental and social governance
 - How responsible is the intervention with regard to environmental and social governance?

5 Unsuccessful

Even unsuccessful innovations have contributed substantially to what came after them

- Some of them were sold to competitors, some failed to keep innovating, some were caught by law

6 Diffusion of innovations

Innovation =

- A (new) idea or practice,
- a (new) product,
- a (new) technique or technology

6.1 Acceptance of innovations

One broad and popular framework:

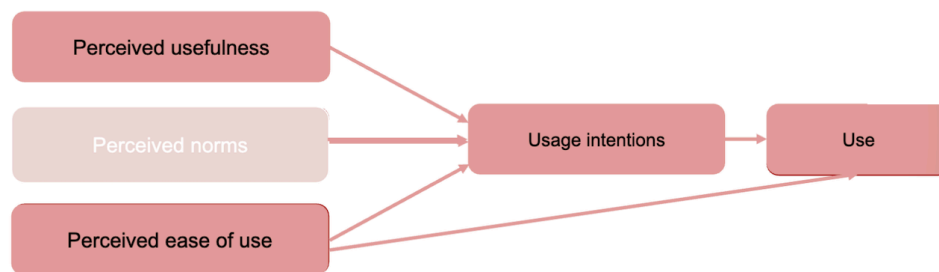


Figure 2: Technology acceptance model by Fred Davis, Rooted in the theory of planned behaviour (TPB) by **Ajzen & Fishbein**

Acceptance as a result of...

- Expectations regarding the performance of a technology (is it doing what it is supposed to?)
- Expectations regarding the (necessary) effort (what do I have to do)?
- Plus: social influence factors (Should I do that? Does everyone do it?)
- New: Further facilitating factors (e.g., trust)

6.2 The role of trust

- Trust = a risky action, a trustee (a subject/person) is willing to risk an object, person, or process that he or she cannot control
- With full control, trust becomes unnecessary
- Central question: Whom and what can I trust?
- Competence, benevolence, and reputation → trustworthiness indicators (AI507 - AI and society/Lectures/3 - Social Cognition)

6.3 Adaption is also a cognitive process

Basic idea: Technology = cognitive construct that requires encoding, storage and retrieval

If technologies are perceived to be similar to familiar or compatible with existing technologies, or related to positive concepts (my family & friends), they are more likely to be perceived as being useful